

Sujith Manikandan

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SKILLS

Programming Languages: Python, Java, C++, JavaScript, TypeScript, HTML, CSS, PHP, R, C#, Erlang

Machine Learning: PyTorch, TensorFlow, Keras, NLTK, Numpy, Pandas, OpenCV, Matplotlib, scikit-learn, Tableau, Hugging Face, Google Colab, Jupyter Notebook, Kaggle, MLFlow

Software: Flask, FastAPI, React, Express, Node.js, Docker, Kubernetes, Git, GitLabs, REST API, Selenium, VMWare, Linux, VS Code

Database and Cloud Technologies: MySQL, PostgreSQL, GraphQL, MongoDB, Firebase, Azure ML, Google Cloud Platform, AWS

Area of Expertise: Natural Language Processing, Computer Vision, Machine Learning, Deep Learning, Artificial Intelligence, Generative AI, MLOps, Finetuning, LLMs

EDUCATION

Concordia University

Sep 2024 – Expected Aug 2026

Master of Applied Computer Science, GPA: 4.15/4.30

Montreal, Canada

Vellore Institute of Technology

Sep 2020 – May 2024

Bachelor of Technology in Computer Science, GPA: 8.88/10

Chennai, India

WORK EXPERIENCE

Research Intern (University of Technology Sydney) [Link to the paper](#)

Aug 2023 – Jan 2024

Argumentation Analysis On Twitter and YouTube Comments Using Ensemble Learning Of Transformers

- Co-authored a research paper on multilingual argumentation analysis, analyzing 3,480+ tweets and YouTube comments in English, Tamil, and code-mixed text for nine discourse attributes like stance, relevancy, and tone.
- Developed and fine-tuned transformer-based NLP models (mBERT, XLM RoBERTa Large, XLM-MLM-100-1280) for multilingual text classification using pyTorch and huggingface, achieving up to 12% higher F1 score than traditional SVM and logistic regression.
- Performed Exploratory Data Analysis (EDA) using NumPy, pandas, matplotlib, and seaborn to uncover key sentiment trends and bias patterns, improving annotation guidelines by 15% agreement in later rounds.
- Designed an ensemble voting mechanism combining majority and single-class voting to boost classification precision, improving misclassification handling for code-mixed data by 10–15%.

PROJECTS

ASR in Indian Languages with enhanced Spell Checker [Link to the paper](#)

Jan 2024 - Apr 2024

PyTorch, MLM, Whisper, GCP, Docker, React, FastAPI, Flask, GIT

- Fine-tuned OpenAI's Whisper-medium ASR model for 4 low-resource Indian languages, achieving an average word error rate reduction from 42% to 35% through advanced error correction.
- Reduced Word Error Rate (WER) by 7% via an ensemble correction module combining XLM-RoBERTa for masked language prediction and Difflib library for fine-grained word-level matching.
- Evaluated robustness under noisy conditions, dialectal variation, and overlapping speech using WER, and CER metrics.
- Executed full training and evaluation pipelines with Hugging Face Transformers and PyTorch, implementing custom tokenization and data preprocessing workflows.
- Developed RESTful APIs for inference using FastAPI and Flask, ensuring flexibility and ease of integration.
- Containerized services with Docker and deployed endpoints on Google Cloud Platform achieving horizontal scalability.
- Applied MLOps best practices by automating CI/CD pipelines and GIT for version control

Semantic Segmentation for Marine species detection Using Custom UNET

[Colab Notebook](#)

PyTorch, UNET, ConvMixer, GCP, Docker, React, FastAPI, Flask

Sep 2024 - Dec 2024

- Designed and built a custom UNet-based semantic segmentation model from scratch in PyTorch for underwater marine species detection in challenging aquatic environments.
- Introduced a novel ConvMixed block (adapted from recent literature) at the UNet's base to enhance feature extraction and improve segmentation accuracy from 78% to 89%. (MIOU metric)
- Built RESTful inference APIs using both FastAPI and Flask, allowing flexible deployment and integration with external applications. Containerized the model and API services using Docker and deployed them to the Google Cloud Platform.